

OK

OMNIKON

NATIONAL HACKATHON 2026

NAMMASHIELD

Multilingual AI Cyber-Safety for Every Indian

Sri Ramakrishna Institute of Technology

ONE MISSION. BUILD THE IMPOSSIBLE.

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TEAM NAME

INNOVATORX

MEMBERS

MALARMATHI A (Individual Participation)

THEME

Cybersecurity — Regional-Language Phishing Detection

PROBLEM STATEMENT

Omni_CyberTech_1 — Regional-Language Phishing Detection. Phishing increasingly targets non-technical users through regional and mixed-language text (Tanglish/Hinglish), a segment existing English-first detectors underserve.

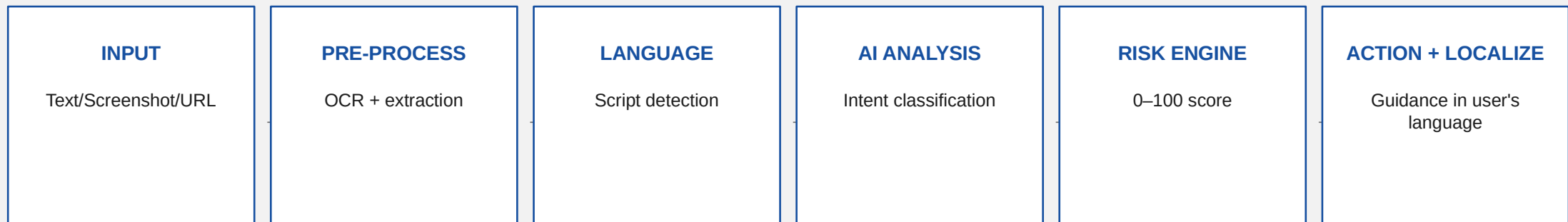
Proposed Solution (Describe your Idea/Solution/Prototype)

- NammaShield is a multilingual AI cyber-safety tool that analyses suspicious text, screenshots, or URLs and returns an explainable 0–100 risk score with localized safety guidance.
- Three-step workflow — DETECT (understand multilingual/mixed-language threats) → EXPLAIN (show evidence behind the score) → PROTECT (recommend the safer next action in the user's language).
- Directly addresses the problem: a generic “phishing detected” label isn't enough for non-technical, regional-language users — they need to understand WHY it's risky and WHAT to do next.
- Handles real mixed-language scam text (e.g. Tanglish KYC/banking phishing) that English-only detectors miss.
- Innovation: integrates multilingual understanding, screenshot OCR, explainable evidence scoring, and localized action guidance into one last-mile cyber-safety workflow — not a new detection algorithm, but a more accessible, trustworthy delivery of protection.
- Demonstrated on both ends of the risk scale: 92/100 on a high-risk KYC phishing SMS vs. 8/100 on a routine delivery notification — showing it discriminates real threats from noise.

Technologies to be used

- Frontend: React + Vite + Tailwind CSS
- Backend: Python + FastAPI | Database: SQLite
- NLP / AI: Pretrained multilingual model(s) + lightweight classification / semantic analysis
- OCR: Tesseract OCR (or comparable lightweight OCR)
- URL Analysis: Python URL parsing + suspicious-pattern indicators

Methodology and process for implementation



Analysis of feasibility

- Uses pretrained multilingual models with lightweight adaptation — not training a transformer from scratch, keeping compute realistic.
- Local machine is sufficient for frontend, backend, SQLite, API development, and lightweight OCR; heavier NLP inference can offload to free-tier cloud.

Potential challenges and risks

- Full language coverage cannot be guaranteed before testing — accuracy varies across languages and mixed-language slang.
- Risk of over- or under-flagging messages if evidence indicators aren't well-calibrated.
- Local RAM/compute limits for larger NLP models.

Strategies to overcome these challenges

- Scope to selected major Indian languages first; expand only after testing (architecture is modular).
- Hybrid decision layer — combine AI output with transparent evidence indicators instead of presenting model output as certainty.
- Use controlled, representative test cases for the demo and keep evaluation data separate from training data.

Potential impact on the target audience

- Non-technical smartphone users who can't easily judge whether a message is genuine.
- Elderly and first-time digital-service users navigating banking/KYC apps for the first time.
- Rural and semi-urban users, and anyone more comfortable in regional or mixed-language chat than English.

Benefits of the solution

- Social: reduces financial and emotional harm from phishing scams by closing the language gap in cyber-safety tools.
- Economic: helps prevent monetary loss from KYC, banking, and OTP scams among vulnerable, underserved users.
- Trust & literacy: explainable evidence builds digital trust rather than just issuing an unexplained warning.
- Scalability: same DETECT → EXPLAIN → PROTECT workflow can extend to job, investment, delivery, and impersonation scams, and to more Indian languages — a genuinely national last-mile digital-safety layer.

Details / Links of the reference and research work

- Omnicon National Hackathon 2026 — Problem Statement Book, Problem Omni_CyberTech_1 (Regional-Language Phishing Detection).
- Real-world mixed-language phishing SMS patterns common in India (banking/KYC-style scam text) used as illustrative reference cases.
- Pretrained multilingual NLP model families (e.g. multilingual transformer-based language models) as the basis for lightweight adaptation rather than training from scratch.
- Tesseract OCR — open-source OCR engine documentation, referenced for the screenshot-scanning pipeline.
- General cybersecurity best practices for phishing indicators — urgency language, credential/OTP requests, suspicious URL structures — used to design the evidence engine.